



Space Technology and Applications

Course code: ET375TGL

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Unit-1

Orbits and Launching Methods: Kepler's Laws, Orbital Elements and Perturbations, Inclined Orbits Antenna Look Angles, Earth Eclipse of Satellite, Sun Transit Outage.

Launch Vehicles: Rocketry, Propellants, Propulsion, Combustion, Solid, Liquid and Cryogenic engines, Control and Guidance system, Ion propulsion and Nuclear Propulsion.



Orbit and Trajectory

- Trajectory is a path traced by a moving body, an orbit is a trajectory that is periodically repeated.
- Motion of an artificial satellite around Earth is an orbit, the path followed by a launch vehicle is a trajectory called the launch trajectory.

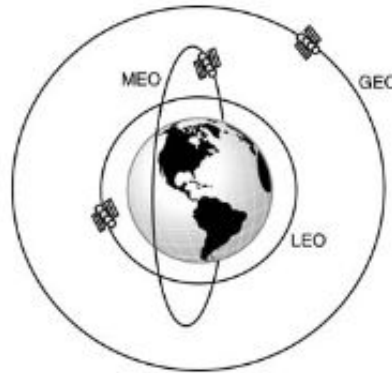


Figure 2.1 Example of orbital motion – satellites revolving around Earth



- The term ‘trajectory’, is associated with a path that is not periodically revisited.
- The path followed by a rocket on its way to the right position for a satellite launch or the path followed by orbiting satellites when they move from an intermediate orbit to their final destined orbit

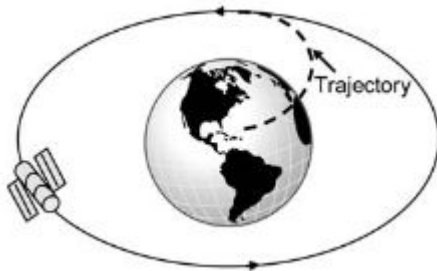


Figure 2.2 Example of trajectory – path followed by a rocket on its way during satellite launch

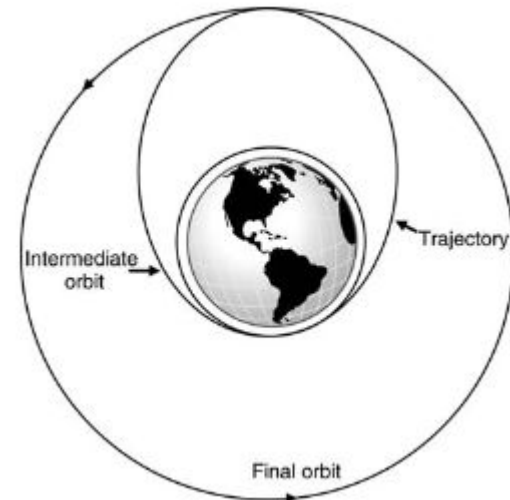


Figure 2.3 Example of trajectory – motion of a satellite from the intermediate orbit to the final orbit



Orbiting Satellites – Basic Principles

- Motion of natural and artificial satellites around Earth is governed by two forces - centripetal force, centrifugal force.
- Centripetal force directed towards the centre of the Earth due to the gravitational force of attraction of Earth. Centrifugal force that acts outwards from the centre of the Earth

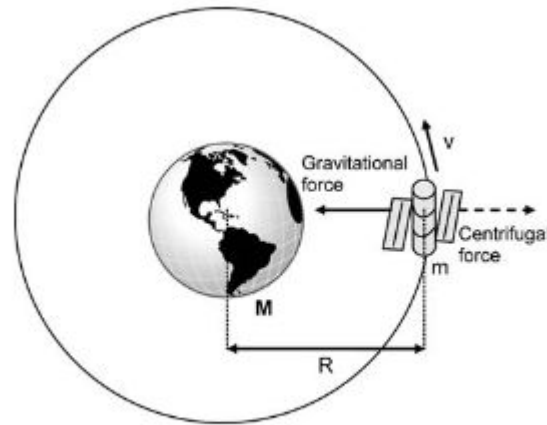


Figure 2.4 Gravitational force and the centrifugal force acting on bodies orbiting Earth



Newton's Law of Gravitation

According to Newton's law of gravitation, every particle irrespective of its mass attracts every other particle with a gravitational force whose magnitude is directly proportional to the product of the masses of the two particles and inversely proportional to the square of the distance between them and written as

$$F = \frac{Gm_1m_2}{r^2}$$

m_1, m_2 = masses of the two particles

r = distance between the two particles

G = gravitational constant = $6.67 \times 10^{-11} \text{ m}^3/\text{kg s}^2$

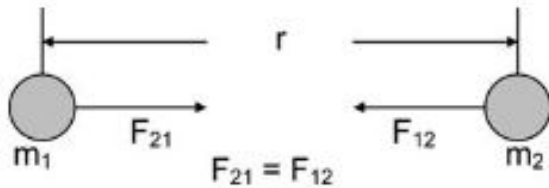


Figure 2.5 Newton's law of gravitation

The forces are equal in magnitude but opposite in direction. The acceleration, which is force per unit mass, experienced by the two particles, however, would depend upon their masses. A larger mass experiences lesser acceleration.



Newton's Second Law of Motion

- According to Newton's second law of motion, the force equals the product of mass and acceleration.
- In the case of a satellite orbiting Earth, if the orbiting velocity is v , then the acceleration, called centripetal acceleration, experienced by the satellite at a distance r from the centre of the Earth would be v^2/r .
- If the mass of satellite is m , it would experience a reaction force of mv^2/r . This is the centrifugal force directed outwards from the centre of the Earth and for a satellite is equal in magnitude to the gravitational force.
- If the satellite orbited Earth with a uniform velocity v , which would be the case when the satellite orbit is a circular one, then equating the two forces mentioned above would lead to an expression for the orbital velocity v as follows:

$$\frac{Gm_1m_2}{r^2} = \frac{m_2v^2}{r}$$
$$v = \sqrt{\left(\frac{Gm_1}{r}\right)} = \sqrt{\left(\frac{\mu}{r}\right)}$$

where

m_1 = mass of Earth

m_2 = mass of the satellite

$$\mu = Gm_1 = 3.986013 \times 10^5 \text{ km}^3/\text{s}^2 = 3.986013 \times 10^{14} \text{ N m}^2/\text{kg}$$



The orbital period in such a case can be computed from

$$T = \frac{2\pi r^{3/2}}{\sqrt{\mu}}$$

In the case of an elliptical orbit, the forces governing the motion of the satellite are the same. The velocity at any point on an elliptical orbit at a distance d from the centre of the Earth is given by the formula

$$v = \sqrt{\left[\mu \left(\frac{2}{d} - \frac{1}{a} \right) \right]}$$

Where, a = semi-major axis of the elliptical orbit

The orbital period in the case of an elliptical orbit is given by

$$T = \frac{2\pi a^{3/2}}{\sqrt{\mu}}$$

The movement of a satellite in an orbit is governed by three Kepler's laws,



Kepler's Laws

Johannes Kepler, based on his lifetime study, gave a set of three empirical expressions that explained planetary motion. These laws were later vindicated when Newton gave the law of gravitation. Though given for planetary motion, these laws are equally valid for the motion of natural and artificial satellites around Earth or for any body revolving around another body. Here, these laws will be discussed with reference to the motion of artificial satellites around Earth.



Kepler's First Law

- The orbit of a satellite around Earth is elliptical with the centre of the Earth lying at one of the foci of the ellipse

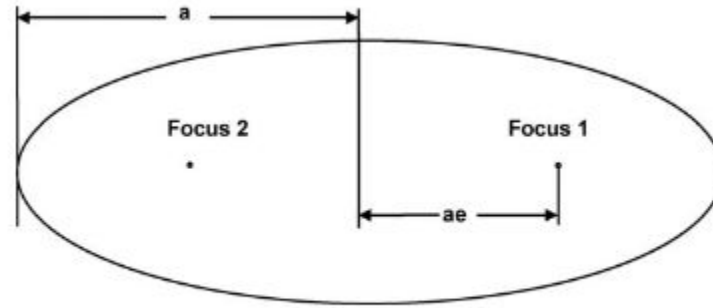


Figure 2.6 Kepler's first law

The elliptical orbit is characterized by its semi-major axis a and eccentricity e . Eccentricity is the ratio of the distance between the centre of the ellipse and either of its foci ($= ae$) to the semi-major axis of the ellipse a .



- A circular orbit is a special case of an elliptical orbit where the foci merge together to give a single central point and the eccentricity becomes zero.
- Important parameters of an elliptical satellite orbit include its **apogee** (farthest point of the orbit from the Earth's centre) and **perigee** (nearest point of the orbit from the Earth's centre) distances.
- For any elliptical motion, the law of conservation of energy is valid at all points on the orbit. The law of conservation of energy states that energy can neither be created nor destroyed; it can only be transformed from one form to another.
- In the context of satellites, it means that the sum of the kinetic and the potential energy of a satellite always remain constant. The value of this constant is equal to $-Gm_1m_2/(2a)$, where, m_1 = mass of Earth, m_2 = mass of the satellite, a = semi-major axis of the orbit



The kinetic and potential energies of a satellite at any point at a distance r from the centre of the Earth are given by

$$\begin{aligned}\text{Kinetic energy} &= \frac{1}{2}(m_2 v^2) \\ \text{Potential energy} &= -\frac{Gm_1 m_2}{r}\end{aligned}$$

Therefore,

$$\begin{aligned}\frac{1}{2}(m_2 v^2) - \frac{Gm_1 m_2}{r} &= -\frac{Gm_1 m_2}{2a} \\ v^2 &= Gm_1 \left(\frac{2}{r} - \frac{1}{a} \right) \\ v &= \sqrt{\left[\mu \left(\frac{2}{r} - \frac{1}{a} \right) \right]}\end{aligned}$$



Kepler's Second Law

line joining the satellite and the centre of the Earth sweeps out equal areas in the plane of the orbit in equal time intervals; i.e. the rate (dA/dt) at which it sweeps area A is constant.

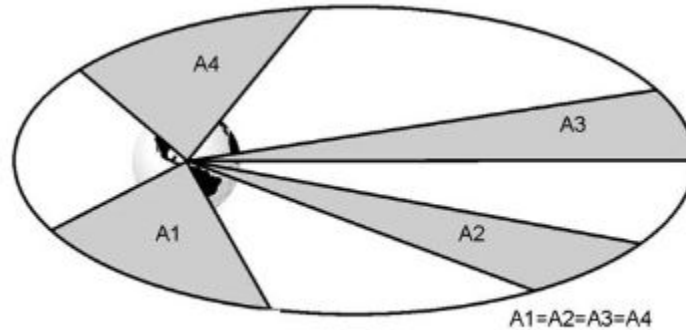


Figure 2.7 Kepler's second law

The rate of change of the swept-out area is given by

$$\frac{dA}{dt} = \frac{\text{angular momentum of the satellite}}{2m}$$

where m is the mass of the satellite.



- Kepler's second law is also equivalent to the law of conservation of momentum, which implies that the angular momentum of the orbiting satellite given by the product of the radius vector and the component of linear momentum perpendicular to the radius vector is constant at all points on the orbit.
- The angular momentum of the satellite of mass m is given by $mr^2\omega$, where ω is the angular velocity of the satellite.
- This further implies that the product $mr^2\omega = (m\omega r)(r) = mv'r$ remains constant.
- Here v' is the component of the satellite's velocity v in the direction perpendicular to the radius vector and is expressed as $v \cos\gamma$, where γ is the angle between the direction of motion of the satellite and the local horizontal, which is in the plane perpendicular to the radius vector r .

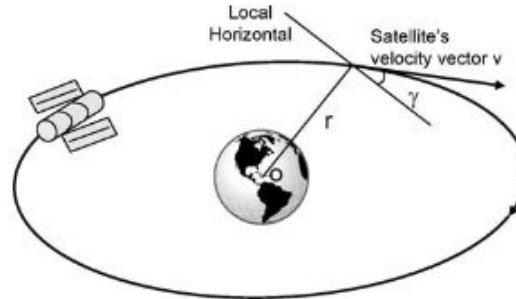


Figure 2.8 Satellite's position at any given time
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- This leads to the conclusion that the product $rv \cos \gamma$ is constant.
- The product reduces to rv in the case of circular orbits and also at apogee and perigee points in the case of elliptical orbits due to angle γ becoming zero.
- It is interesting to note here that the velocity component v' is inversely proportional to the distance r .
- Qualitatively, this implies that the satellite is at its lowest speed at the apogee point and the highest speed at the perigee point.
- In other words, for any satellite in an elliptical orbit, the dot product of its velocity vector and the radius vector at all points is constant. Hence,

$$v_p r_p = v_a r_a = vr \cos \gamma$$

where

v_p = velocity at the perigee point

r_p = perigee distance

v_a = velocity at the apogee point

r_a = apogee distance

v = satellite velocity at any point in the orbit

r = distance of the point

γ = angle between the direction of motion of the satellite and the local horizontal



Kepler's Third Law

- According to the Kepler's third law, also known as the law of periods, the square of the time period of any satellite is proportional to the cube of the semi-major axis of its elliptical orbit.
- The expression for the time period can be derived as follows. A circular orbit with radius r is assumed. Remember that a circular orbit is only a special case of an elliptical orbit with both the semi-major axis and semi-minor axis equal to the radius. Equating the gravitational force with the centrifugal force gives

$$\frac{Gm_1m_2}{r^2} = \frac{m_2v^2}{r}$$

- Replacing v by ωr in the above equation gives

$$\frac{Gm_1m_2}{r^2} = \frac{m_2\omega^2r^2}{r} = m_2\omega^2r$$

which gives $\omega^2 = Gm_1/r^3$



Substituting $\omega = 2\pi/T$ gives

$$T^2 = \left(\frac{4\pi^2}{Gm_1} \right) r^3$$

This can also be written as

$$T = \left(\frac{2\pi}{\sqrt{\mu}} \right) r^{3/2}$$

The above equation holds good for elliptical orbits provided r is replaced by the semi-major axis a . This gives the expression for the time period of an elliptical orbit as

$$T = \left(\frac{2\pi}{\sqrt{\mu}} \right) a^{3/2}$$



Orbital Parameters

- Ascending and descending nodes
- Equinoxes
- Solstices
- Apogee
- Perigee
- Eccentricity
- Semi-major axis
- Right ascension of the ascending node
- Inclination
- Argument of the perigee
- True anomaly of the satellite
- Angles defining the direction of the satellite



Ascending and descending nodes.

- The satellite orbit cuts the equatorial plane at two points: the first, called the descending node (N1), where the satellite passes from the northern hemisphere to the southern hemisphere, and the second, called the ascending node (N2), where the satellite passes from the southern hemisphere to the northern hemisphere

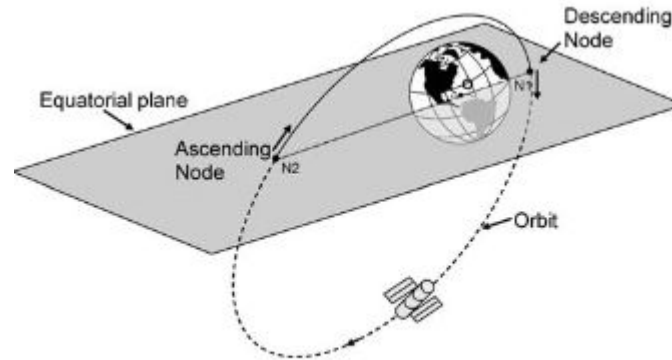


Figure 2.9 Ascending and descending nodes



Equinoxes.

- The inclination of the equatorial plane of Earth with respect to the direction of the sun, defined by the angle formed by the line joining the centre of the Earth and the sun with the Earth's equatorial plane follows a sinusoidal variation and completes one cycle of sinusoidal variation over a period of 365 days

$$\text{Inclination angle (in degrees)} = 23.4 \sin \left(\frac{2\pi t}{T} \right)$$

During the equinoxes, it can be seen that the equatorial plane of Earth will be aligned with the direction of the sun.

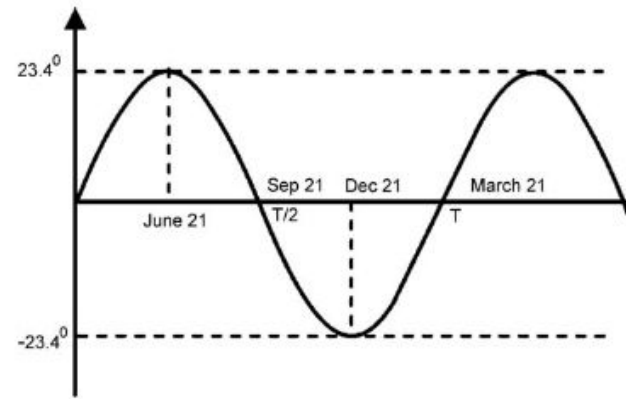


Figure 2.10 Yearly variation of angular inclination of Earth with the sun



Also, the line of intersection of the Earth's equatorial plane and the Earth's orbital plane that passes through the centre of the Earth is known as the line of equinoxes. The direction of this line with respect to the direction of the sun on 20-21 March determines a point at infinity called the vernal equinox (Y)

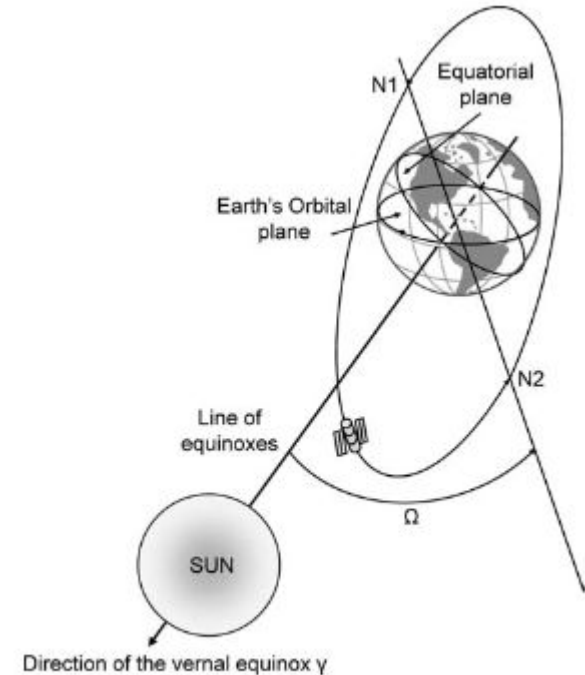


Figure 2.11 Vernal equinox



Solstices

- Solstices are the times when the inclination angle is at its maximum, i.e. 23.4° . These also occur twice during a year on 20-21 June, called the summer solstice, and 21-22 December, called the winter solstice.



Apogee.

Apogee is the point on the satellite orbit that is at the farthest distance from the centre of the Earth

$$\text{Apogee distance} = a(1 + e)$$

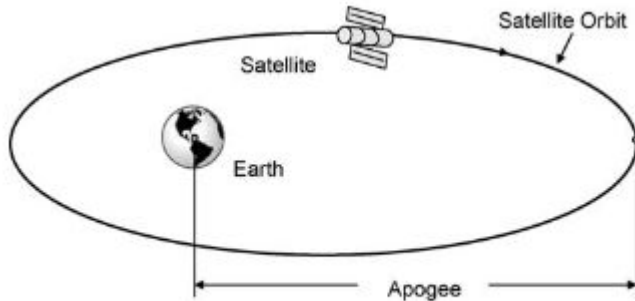


Figure 2.12 Apogee

The apogee distance can also be computed from the known values of the perigee distance and velocity at the perigee V_p from

$$V_p = \sqrt{\left(\frac{2\mu}{\text{Perigee distance}} - \frac{2\mu}{\text{Perigee distance} + \text{Apogee distance}} \right)}$$

where

$$V_p = V \left(\frac{d \cos \gamma}{\text{Perigee distance}} \right)$$



Perigee.

- Perigee is the point on the orbit that is nearest to the centre of the Earth

$$\text{Perigee distance} = a(1 - e)$$

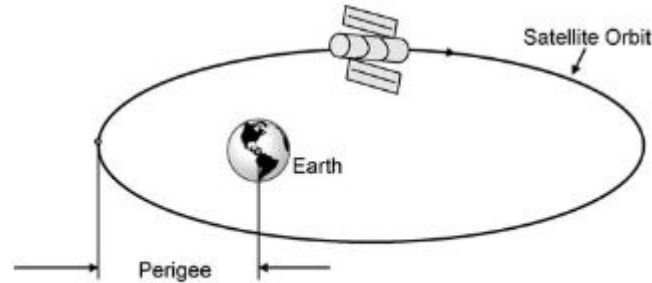


Figure 2.13 Perigee



Eccentricity.

- The orbit eccentricity e is the ratio of the distance between the centre of the ellipse and the centre of the Earth to the semi-major axis of the ellipse.

$$e = \frac{\text{apogee} - \text{perigee}}{\text{apogee} + \text{perigee}}$$

$$e = \frac{\text{apogee} - \text{perigee}}{2a}$$

Semi-major axis.

$$a = \frac{\text{apogee} + \text{perigee}}{2}$$



Right ascension of the ascending node.

- The right ascension of the ascending node tells about the orientation of the line of nodes, which is the line joining the ascending and descending nodes, with respect to the direction of the vernal equinox. It is expressed as an angle Ω measured from the vernal equinox towards the line of nodes in the direction of rotation of Earth. The angle could be anywhere from 0° to 360° .

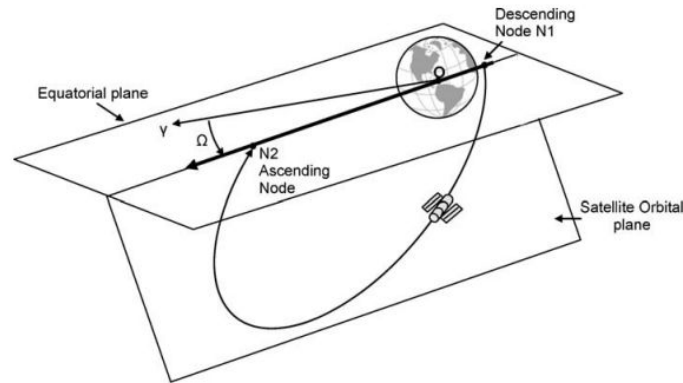


Figure 2.14 Right ascension of the ascending node



- Acquisition of the correct angle of right ascension of the ascending node (Ω) is important to ensure that the satellite orbits in the given plane.
- This can be achieved by choosing an appropriate injection time depending upon the longitude.
- Angle β can be computed as the difference between two angles. One is the angle α between the direction of the vernal equinox and the longitude of the injection point and the other is the angle β between the line of nodes and the longitude of the injection point,

Angle β can be computed from

$$\sin \beta = \frac{\cos i \sin l}{\cos l \sin i}$$

where i is the orbit inclination and l is the latitude at the injection point.

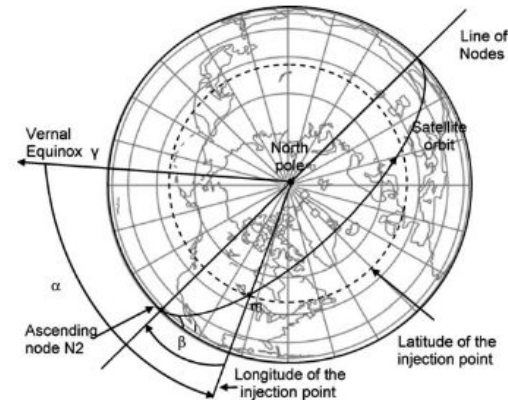


Figure 2.15 Computation of the right ascension of the ascending node



Inclination

- Inclination is the angle that the orbital plane of the satellite makes with the Earth's equatorial plane.

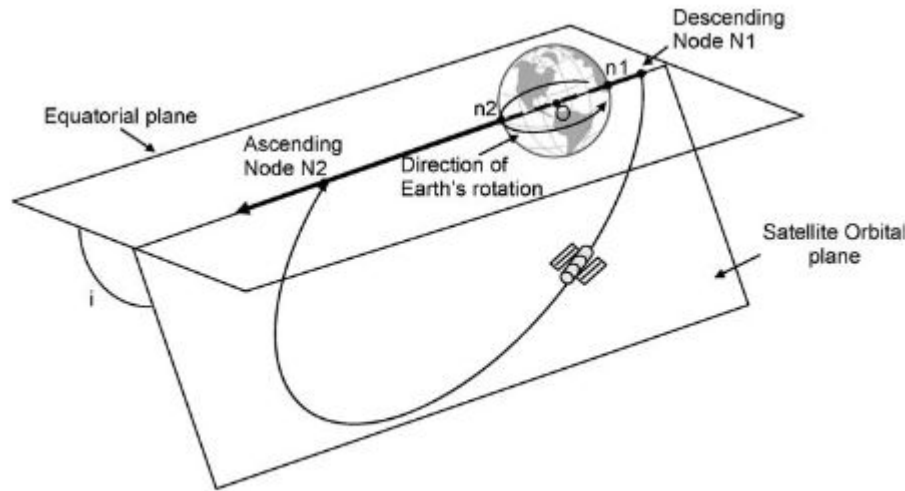


Figure 2.16 Angle of inclination

$$\cos i = \sin A_z \cos l$$

It is measured as follows. The line of nodes divides both the Earth's equatorial plane as well as the satellite's orbital plane into two halves. Inclination is measured as the angle between that half of the satellite's orbital plane containing the trajectory of the satellite from the descending node to the ascending node to that half of the Earth's equatorial plane containing the trajectory of a point on the equator from $n1$ to $n2$, where $n1$ and $n2$ are respectively the points vertically below the descending and ascending nodes (Figure 2.16). The inclination angle can be determined from the latitude l at the injection point and the angle A_z between the projection of the satellite's velocity vector on the local horizontal and north.



Argument of the perigee.

- This parameter defines the location of the major axis of the satellite orbit.
- It is measured as the angle ω between the line joining the perigee and the centre of the Earth and the line of nodes from the ascending node to the descending node in the same direction as that of the satellite orbit

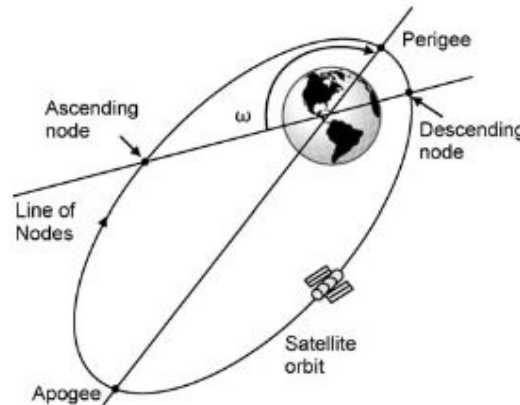


Figure 2.17 Argument of perigee



True anomaly of the satellite.

- This parameter is used to indicate the position of the satellite in its orbit.
- This is done by defining an angle θ , called the true anomaly of the satellite, formed by the line joining the perigee and the centre of the Earth with the line joining the satellite and the centre of the Earth

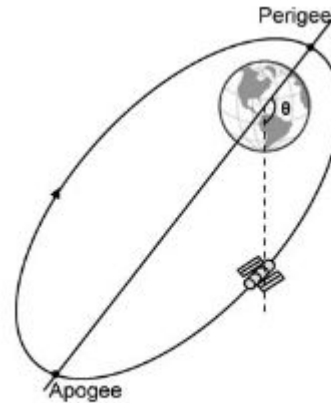


Figure 2.18 True anomaly of a satellite



Angles defining the direction of the satellite.

- The direction of the satellite is defined by two angles, the first by angle γ between the direction of the satellite's velocity vector and its projection in the local horizontal and the second by angle A_z between the north and the projection of the satellite's velocity vector on the local horizontal

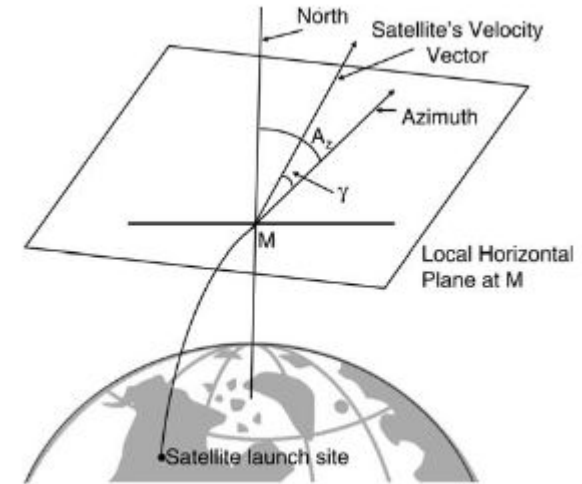


Figure 2.19 Angles defining the direction of the satellite



Problem 2.1

A satellite is orbiting Earth in a uniform circular orbit at a height of 630 km from the surface of Earth. Assuming the radius of Earth and its mass to be 6370 km and 5.98×10^{24} kg respectively, determine the velocity of the satellite (Take the gravitational constant $G = 6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2$).

Solution:

Orbit radius $R = 6370 + 630 = 7000 \text{ km} = 7\,000\,000 \text{ m}$

Also, constant $\mu = GM = 6.67 \times 10^{-11} \times 5.98 \times 10^{24}$
 $= 39.8 \times 10^{13} \text{ N m}^2/\text{kg}$
 $= 39.8 \times 10^{13} \text{ m}^3/\text{s}^2$

The velocity of the satellite can be computed from

$$v = \sqrt{\left(\frac{\mu}{R}\right)} = \sqrt{\left(\frac{39.8 \times 10^{13}}{7\,000\,000}\right)} = 7.54 \text{ km/s}$$



Problem 2.2

The apogee and perigee distances of a satellite orbiting in an elliptical orbit are respectively 45 000 km and 7000 km. Determine the following:

1. Semi-major axis of the elliptical orbit
2. Orbit eccentricity
3. Distance between the centre of the Earth and the centre of the elliptical orbit

Solution:

$$\begin{aligned} 1. \text{ Semi-major axis of the elliptical orbit } a &= \frac{\text{apogee} + \text{perigee}}{2} \\ &= \frac{45\,000 + 7\,000}{2} = 26\,000 \text{ km} \end{aligned}$$

$$2. \text{ Eccentricity } e = \frac{\text{apogee} - \text{perigee}}{2a} = \frac{45\,000 - 7\,000}{2 \times 26\,000} = \frac{38\,000}{52\,000} = 0.73$$

$$\begin{aligned} 3. \text{ Distance between the centre of the Earth and the centre of the ellipse} &= ae \\ &= 26\,000 \times 0.73 \\ &= 18\,980 \text{ km} \end{aligned}$$



A satellite is moving in an elliptical orbit with the major axis equal to 42,000 km. If the perigee distance is 8,000 km, find the apogee and the orbit eccentricity.

Solution:

Major axis = 42 000 km

Also, major axis = apogee + perigee = 42 000 km

Therefore apogee = 42 000 – 8000 = 34 000 km

Also, eccentricity $e = \frac{\text{apogee} - \text{perigee}}{\text{major axis}}$

$$= \frac{34\,000 - 8\,000}{42\,000}$$

$$= \frac{26\,000}{42\,000} = 0.62$$



Problem 2.4

Refer to the satellite orbit of Figure 2.20. Determine the apogee and perigee distances if the orbit eccentricity is 0.6.

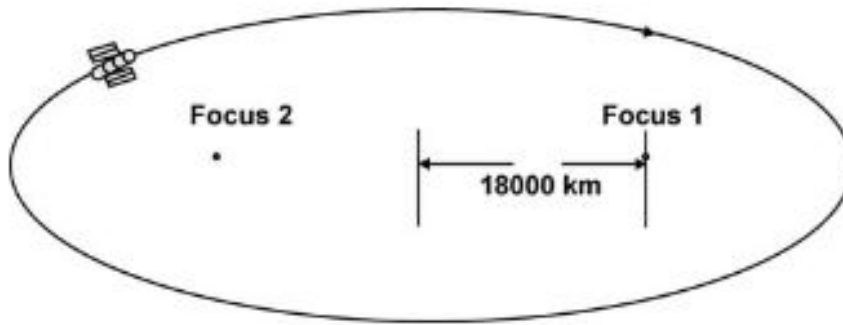


Figure 2.20 Figure for Problem 2.4

Solution: If e is the orbit eccentricity and a the semi-major axis of the elliptical orbit, then the distance between the centre of the Earth and the centre of the ellipse is equal to ae .

Therefore $ae = 18\,000\text{ km}$

This gives $a = 18\,000/e = 18\,000/0.6 = 30\,000\text{ km}$

Apogee distance $= a(1 + e) = 30\,000 \times (1 + 0.6) = 48\,000\text{ km}$

Perigee distance $= a(1 - e) = 30\,000 \times (1 - 0.6) = 12\,000\text{ km}$



The difference between the furthest and the closest points in a satellite's elliptical orbit from the surface of the Earth is 30,000 km and the sum of the distances is 50,000 km. If the mean radius of the Earth is considered to be 6,400 km, determine the orbit eccentricity.

Solution: Apogee – Perigee = 30 000 km as the radius of the Earth will cancel in this case

$$\text{Apogee} + \text{Perigee} = 50\,000 + 2 \times 6400 = 62\,800 \text{ km}$$

$$\text{Orbit eccentricity} = (\text{Apogee} - \text{Perigee})/(\text{Apogee} + \text{Perigee}) = 30\,000/62\,800 = 0.478$$



Problem 2.6

Refer to Figure 2.21. Satellite A is orbiting Earth in a near-Earth circular orbit of radius 7000 km. Satellite B is orbiting Earth in an elliptical orbit with apogee and perigee distances of 47 000 km and 7000 km respectively. Determine the velocities of the two satellites at point X. (Take $\mu = 39.8 \times 10^{13} \text{ m}^3/\text{s}^2$.)

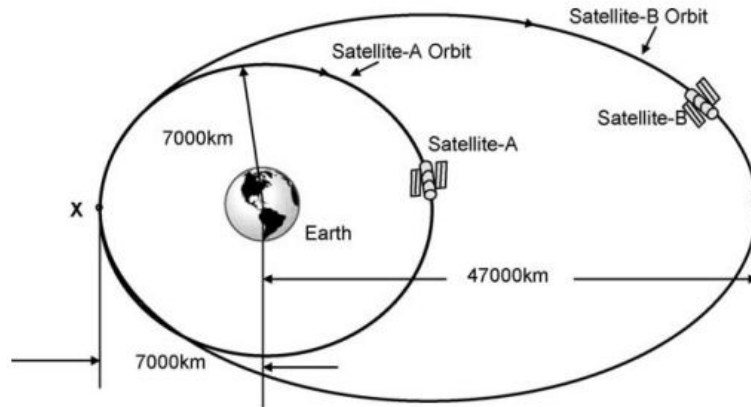


Figure 2.21 Figure for Problem 2.6



Solution: The velocity of a satellite moving in a circular orbit is constant throughout the orbit and is given by

$$v = \sqrt{\left(\frac{\mu}{R}\right)}$$

$$\begin{aligned}\text{Therefore the velocity of satellite A at point X} &= \sqrt{\left(\frac{39.8 \times 10^{13}}{7\,000\,000}\right)} \\ &= 7.54 \text{ km/s}\end{aligned}$$

The velocity of the satellite at any point in an elliptical orbit is given by

$$v = \sqrt{\left[\mu \left(\frac{2}{R} - \frac{1}{a}\right)\right]}$$

where a is the semi-major axis and R is the distance of the point in question from the centre of the Earth. Here $R = 7000$ km and $a = (47\,000 + 7000)/2 = 27\,000$ km. Therefore, velocity of satellite B at point X is given by

$$v = \sqrt{\left[(39.8 \times 10^{13}) \times \left(\frac{2}{7\,000\,000} - \frac{1}{27\,000\,000}\right)\right]} = 9.946 \text{ km/s}$$

**Problem 2.9**

Calculate the orbital period of a satellite moving in an elliptical orbit having a major axis of 50 000 km. (Take $\mu = 39.8 \times 10^{13} \text{ Nm}^2/\text{kg}$).

Solution: Semi-major axis, $a = 50\,000/2 = 25\,000 \text{ km}$

$$\begin{aligned}\text{Orbital time period, } T &= 2\pi \times \sqrt{(a^3/\mu)} = 2 \times 3.14 \times \sqrt{(25\,000\,000)^3/(39.8 \times 10^{13})} \\ &= 39\,344 \text{ seconds} = 10 \text{ hours } 55 \text{ minutes } 44 \text{ seconds}\end{aligned}$$



Problem 2.10

Refer to Figure 2.24. The semi-major axes of the two satellites shown in the figure are 18 000 km (satellite 1) and 24 000 km (satellite 2). Determine the relationship between their orbital periods.

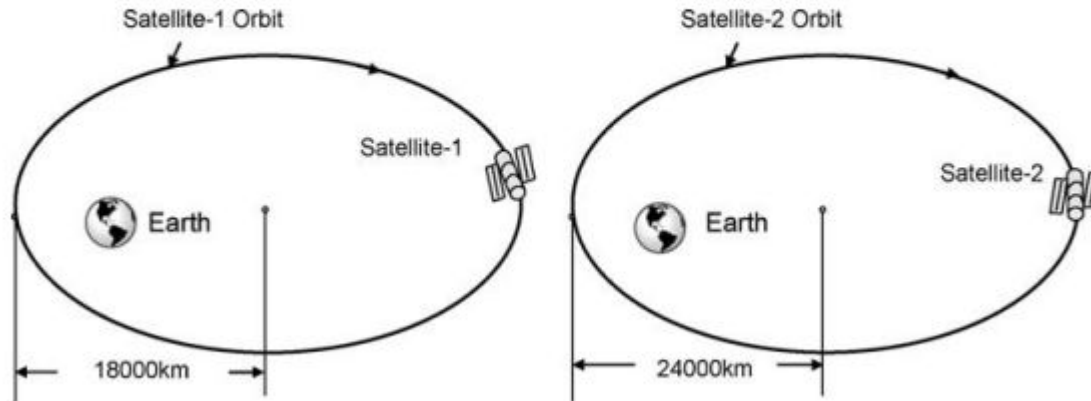


Figure 2.24 Figure for Problem 2.10



Solution: The orbital time period $T = 2\pi \times \sqrt{a^3/\mu}$ where a is the semi-major axis. If a_1 and a_2 are the semi-major axes in the two cases and T_1 and T_2 are their time periods, then

$$T_1 = 2\pi \times \sqrt{\left(\frac{a_1^3}{\mu}\right)}$$

and

$$T_2 = 2\pi \times \sqrt{\left(\frac{a_2^3}{\mu}\right)}$$

This gives

$$\frac{T_2}{T_1} = \left(\frac{a_2}{a_1}\right)^{3/2} = \left(\frac{24\,000}{18\,000}\right)^{3/2} = 1.54$$

The orbital period of satellite 2 is 1.54 times that of satellite 1.



Problem 2.11

The elliptical eccentric orbit of a satellite has its semi-major and semi-minor axes as 25 000 km and 18 330 km respectively. Determine the apogee and perigee distances.

Solution: If r_a and r_p are the apogee and perigee distances respectively, a and b are the semi-major and semi-minor axis respectively and e is the eccentricity, then

$$a = \frac{r_a + r_p}{2} \quad b = \sqrt{(r_a r_p)}$$

Now $(r_a + r_p)/2 = 25\,000$, which gives $r_a + r_p = 50\,000$

and $\sqrt{(r_a r_p)} = 18\,330$, which gives $r_a r_p = 335\,988\,900$

Substituting the value of r_p from these equations gives

$$r_a(50\,000 - r_a) = 335\,988\,900$$

which gives

$$r_a^2 - 50\,000r_a + 335\,988\,900 = 0$$

Solving this quadratic equation gives two solutions:

$$r_a = 42\,000 \text{ km, } 8000 \text{ km}$$

If the semi-major axis is 25 000 km, r_a cannot be 8000 km. Therefore

$$\text{Apogee distance } r_a = 42\,000 \text{ km}$$

and

$$\text{Perigee distance } r_p = 50\,000 - 42\,000 = 8000 \text{ km}$$

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Alternative solution:

$$e = \frac{\sqrt{(a^2 - b^2)}}{a} = e = \frac{\sqrt{[(25\,000)^2 - (18\,330)^2]}}{25\,000} = 0.68$$

This gives

Apogee distance = $a(1 + e) = 25\,000 \times 1.68 = 42\,000$ km

and Perigee distance = $a(1 - e) = 25\,000 \times 0.32 = 8\,000$ km

Therefore, from both the methods,

Apogee distance = 42 000 km and Perigee distance = 8000 km



Injection Velocity and Resulting Satellite Trajectories

- The horizontal velocity with which a satellite is injected into space by the launch vehicle with the intention of imparting a specific trajectory to the satellite has a direct bearing on the satellite trajectory.
- The phenomenon is best explained in terms of the three cosmic velocities.
- The general expression for the velocity of a satellite at the perigee point (VP), assuming an elliptical orbit, is given by

$$V_P = \sqrt{\left[\left(\frac{2\mu}{r}\right) - \left(\frac{2\mu}{R+r}\right)\right]}$$

where

R = apogee distance

r = perigee distance

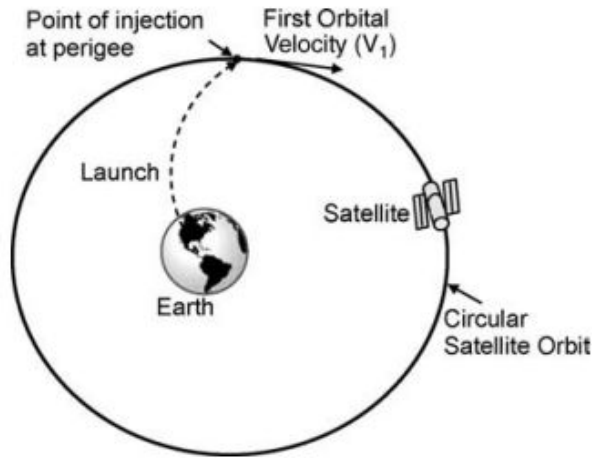
$\mu = GM = \text{constant}$

- The first cosmic velocity V_1 is the one at which apogee and perigee distances are equal, i.e. $R = r$, and the orbit is circular. The above expression then reduces to

$$V_1 = \sqrt{\left(\frac{\mu}{r}\right)}$$



- Thus, irrespective of the distance r of the satellite from the centre of the Earth, if the injection velocity is equal to the first cosmic velocity, also sometimes called the first orbital velocity, the satellite follows a circular orbit and moves with a uniform velocity equal to $\sqrt{(\mu/r)}$.



A simple calculation shows that for a satellite at 35 786 km above the surface of the Earth, the first cosmic velocity turns out to be 3.075 km/s and the orbital period is 23 hours 56 minutes, which is equal to the time period of one sidereal day – the time taken by Earth to complete one full rotation around its axis with reference to distant stars.

Figure 2.27 Satellite's path where the injection velocity is equal to the first orbital velocity



- If the injection velocity happens to be less than the first cosmic velocity, the satellite follows a ballistic trajectory and falls back to Earth.
- In fact, in this case, the orbit is elliptical and the injection point is at the apogee and not the perigee. If the perigee lies in the atmosphere or exists only virtually below the surface of the Earth, the satellite accomplishes a ballistic flight and falls back to Earth.

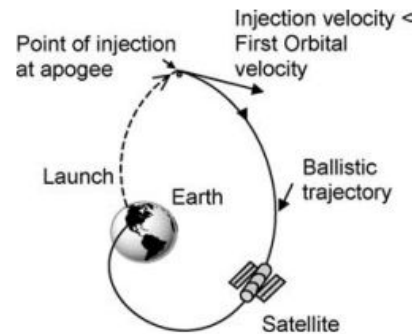


Figure 2.28 Satellite's path where the injection velocity is less than the first orbital velocity



- For injection velocity greater than the first cosmic velocity and less than the second cosmic velocity, i.e. $V > \sqrt{\mu/r}$ and $V < \sqrt{2\mu/r}$, the orbit is elliptical and eccentric.
- The orbit eccentricity is between 0 and 1.
- The injection point in this case is the perigee and the apogee distance attained in the resultant elliptical orbit depends upon the injection velocity.
- The higher the injection velocity, the greater is the apogee distance.
- The apogee distance can also be computed from the known value of injection velocity, which is also the velocity at the perigee point as the perigee coincides with the injection point, and the velocity v at any other point in the orbit distant d from the centre of the Earth using

$$V_p = \sqrt{\left[(2\mu/r) - \left(\frac{2\mu}{R+r} \right) \right]} = \frac{vd \cos \gamma}{r}$$



- When the injection velocity equals $\sqrt{2\mu/r}$, the apogee distance R becomes infinite and the orbit takes the shape of a parabola and the orbit eccentricity is 1.
- This is the second cosmic velocity v_2 . At this velocity, the satellite escapes Earth's gravitational pull.

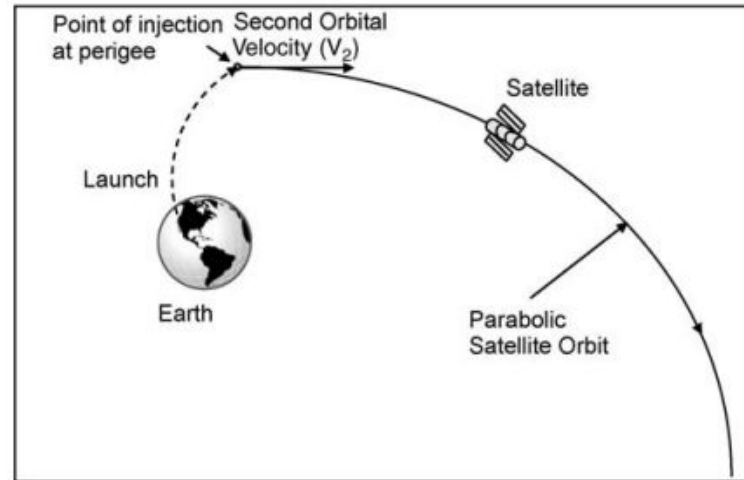


Figure 2.29 Satellite's path where the injection velocity is equal to the second orbital velocity
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- For an injection velocity greater than the second cosmic velocity, the trajectory is hyperbolic within the solar system and the orbit eccentricity is greater than 1.
- If the injection velocity is increased further, a stage is reached where the satellite succeeds in escaping from the solar system.
- This is known as the third cosmic velocity and is related to the motion of planet Earth around the sun.
- The third cosmic velocity (V_3) is mathematically expressed as

$$V_3 = \sqrt{\left[\frac{2\mu}{r} - V_t^2(3 - 2\sqrt{2}) \right]}$$

where, V_t is the speed of Earth's revolution around the sun



- Beyond the third cosmic velocity, there is a region of hyperbolic flights outside the solar system.
- Coming back to elliptical orbits, the greater the injection velocity from the first cosmic velocity, the greater is the apogee distance.
- This is evident from the generalized expression for the velocity of the satellite in elliptical orbits according to which

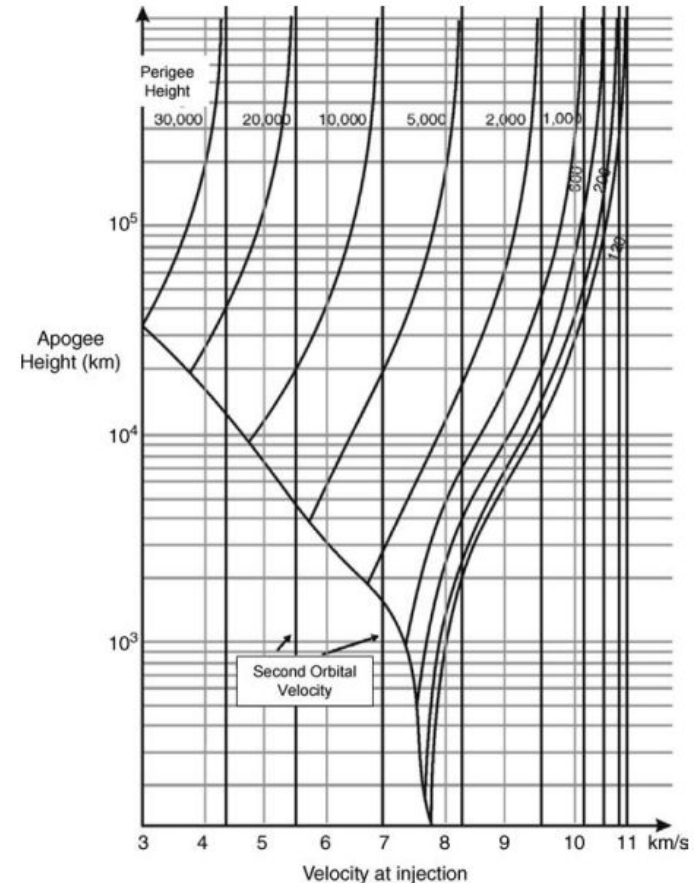
V_p = velocity at the perigee point

$$= \sqrt{\left[(2\mu) \left(\frac{1}{r} - \frac{1}{R+r} \right) \right]}$$

- For a given perigee distance r , a higher velocity at the perigee point, which is also the injection velocity, necessitates that the apogee distance R is greater.



- Figure 2.30 shows a family of curves that can be used to find out the attained apogee height for a given value of injection velocity at the perigee point.
- The dashed line shows the relationship between the injection velocity and altitude for a circular orbit.



Nagendra N N, Dep **Figure 2.30** Family of curves showing the apogee height for various values of the injection velocity



Problem 2.15

A satellite launched with an injection velocity of v_1 from a point above the surface of the Earth at a distance r from the centre of the Earth attains an elliptical orbit with an apogee distance of R_1 . The same satellite when launched with an injection velocity of v_2 from the same perigee distance attains an elliptical orbit with an apogee distance of R_2 . Derive the relationship between v_1 and v_2 in terms of r , R_1 and R_2 .

Solution: Refer to Figure 2.31. Expressions for the velocity at perigee points can be written as

$$v_1 = \sqrt{\left[(2\mu) \left(\frac{1}{r} - \frac{1}{R_1 + r} \right) \right]}$$

$$v_2 = \sqrt{\left[(2\mu) \left(\frac{1}{r} - \frac{1}{R_2 + r} \right) \right]}$$

Squaring the two expressions and then taking the ratio of the two yields

$$\begin{aligned} \left(\frac{v_2}{v_1} \right)^2 &= \frac{(1/r) - 1/(R_2 + r)}{(1/r) - 1/(R_1 + r)} \\ &= \frac{R_2/r(R_2 + r)}{R_1/r(R_1 + r)} \\ &= \left(\frac{R_2}{R_1} \right) \left(\frac{R_1 + r}{R_2 + r} \right) \\ &= \frac{1 + r/R_1}{1 + r/R_2} \end{aligned}$$

Therefore,

$$\left(\frac{v_2}{v_1} \right)^2 = \frac{1 + r/R_1}{1 + r/R_2}$$

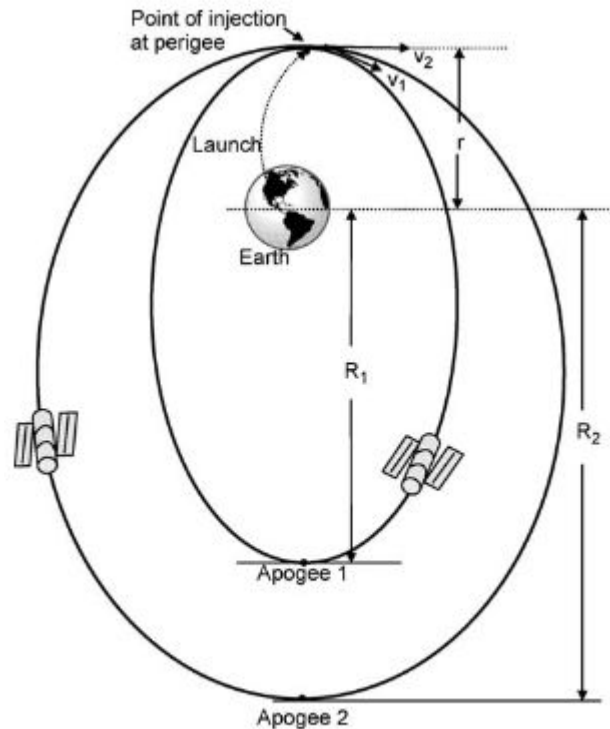


Figure 2.31 Figure for Problem 2.15



Problem 2.16

A satellite launched with an injection velocity of (v_1) from a point above the surface of the Earth at a distance of (r) of 8000 km from the centre of the Earth attains an elliptical orbit with an apogee distance of 12 000 km. The same satellite when launched with an injection velocity of (v_2) that is 20 % higher than (v_1) from the same perigee distance attains an elliptical orbit with an apogee distance of (R_2). Determine the new apogee distance.

Solution:

We know that: $(v_2/v_1)^2 = [(1 + r/R_1)/(1 + r/R_2)]$

Substituting $v_2 = 1.2v_1$, we get $1.44 = [(1 + 8000/12\,000)/(1 + 8000/R_2)]$

or $1.44 \times 3 \times (1 + 8000/R_2) = 5$ or $(1 + 8000/R_2) = 1.1574$

$R_2 = 8000/0.1574 = 50\,826$ km



Problem 2.17

A rocket injects a satellite with a horizontal velocity of 8 km/s from a height of 1620 km from the surface of the Earth. What will be the velocity of the satellite at a point distant 10 000 km from the centre of the Earth, if the direction of the satellite makes an angle of 30° with the local horizontal at that point? Assume radius of the Earth to be 6380 km.

Solution:

We know that $V_2 = \sqrt{\{2\mu/r\} - \{2\mu/(R+r)\}} = (v.d.\cos\gamma)/r$

Here $V_2 = 8$ km/s, $r = 1620 + 6380 = 8000$ km, $d = 10\,000$ km and $\gamma = 30^\circ$

Therefore, $[v \times 10\,000 \times \cos 30^\circ / 8000] = 8$

or $v = (8 \times 8000) / (10\,000 \times 0.866) = 7.39$ km/s



Problem 2.18

A rocket injects a satellite with a certain horizontal velocity from a height of 620 km from the surface of the Earth. The velocity of the satellite at a point distant 9000 km from the centre of the Earth is observed to be 8 km/s. If the direction of the satellite makes an angle of 30° with the local horizontal at that point, determine the apogee distance of the satellite orbit. (Assume that the radius of the Earth is 6380 km and $\mu = 39.8 \times 10^{13} \text{ m}^3/\text{s}^2$.)

Solution:

It is known that
$$v_p = \sqrt{\left[\left(\frac{2\mu}{r}\right) - \left(\frac{2\mu}{R+r}\right)\right]} = \frac{vd \cos \gamma}{r}$$

Here,

$$r = 620 + 6380 = 7000 \text{ km}$$

$$d = 9000 \text{ km}$$

$$v = 8 \text{ km/s}$$

$$\gamma = 30^\circ$$

$$R = \text{apogee distance} = ?$$

Therefore,

$$\sqrt{\left[\left(\frac{2\mu}{r}\right) - \left(\frac{2\mu}{R+r}\right)\right]} = 8 \times 9000 \times \frac{0.866}{7000} = 8.90 \text{ km/s} = 8.90 \times 10^3 \text{ m/s}$$

or

$$\frac{2\mu}{r} - \frac{2\mu}{R+r} = (8.90 \times 10^3)^2 = 79.34 \times 10^6$$

or

$$\frac{2\mu}{R+r} = \frac{2 \times 39.8 \times 10^{13}}{7000000} - (79.34 \times 10^6) = 34.36 \times 10^6$$

This gives

$$R+r = \frac{2 \times 39.8 \times 10^{13}}{34.36 \times 10^6} = 2.317 \times 10^7 \text{ m}$$

$$\text{or } R = (2.317 \times 10^7 - 7000000) \text{ m} = 16170 \text{ km}$$



Types of Satellite Orbits

- The satellite orbits can be classified on the basis of:
 - Orientation of the orbital plane
 - Eccentricity
 - Distance from Earth



Orientation of the Orbital Plane

Orbits can be classified as equatorial orbits, polar orbits and inclined orbits.

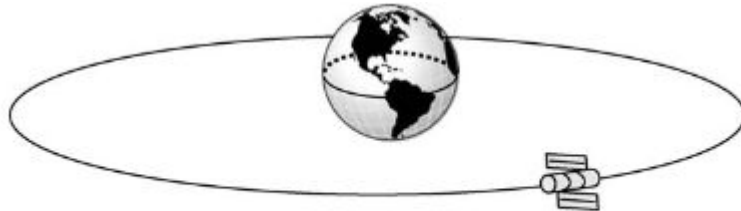


Figure 2.32 Equatorial orbit

Angle of inclination is zero



Figure 2.33 Polar orbit

Angle of inclination
equal to 90°



Angle of inclination between 0° and 180° - Inclined orbit



Figure 2.34 Prograde orbit

Direct or prograde orbit - Inclinations between 0° and 90° , the satellite travels in the same direction as the direction of rotation of the Earth



Figure 2.35 Retrograde orbit

Retrograde orbit - Inclinations between 90° and 180° , the satellite orbits in a direction opposite to the direction of rotation of the Earth



Eccentricity of the Orbit

Orbits are classified as elliptical and circular orbits

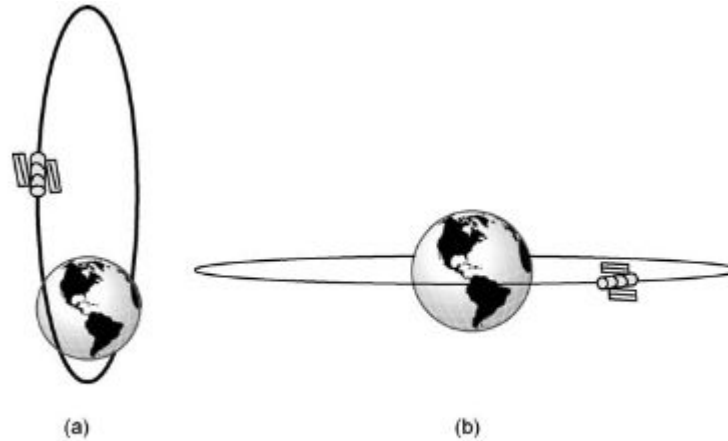


Figure 2.36 (a) Elliptical orbit and (b) circular orbit

Orbit eccentricity lies between 0 and 1, the orbit is elliptical with the centre of the Earth lying at one of the foci of the ellipse. When the eccentricity is zero, the orbit becomes circular.



Distance from Earth

Classified as low Earth orbits (LEOs), medium Earth orbits (MEOs) and geostationary Earth orbits (GEOs),

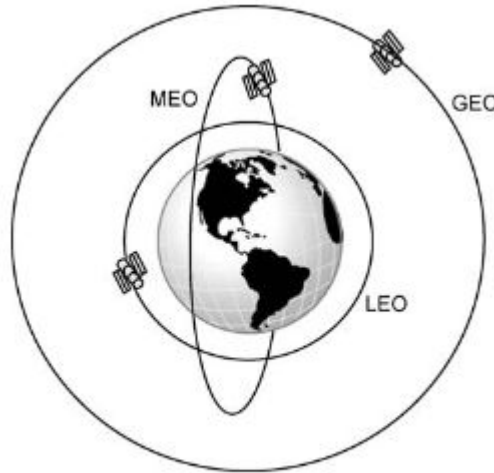


Figure 2.38 LEO, MEO and GEO



Low Earth orbits

- Height of around 160 to 500 km above the surface of the Earth.
- These satellites, being closer to the surface of the Earth, have much shorter orbital periods and smaller signal propagation delays.
- A lower propagation delay makes them highly suitable for communication applications.
- Due to lower propagation paths, the power required for signal transmission is also less, with the result that the satellites are of small physical size and are inexpensive to build.
- However, due to a shorter orbital period, of the order of an hour and a half or so, these satellites remain over a particular ground station for a short time.
- Hence, several of these satellites are needed for 24 hour coverage.



Medium Earth orbit (MEO) satellites orbit

- Distance of approximately 10 000 to 20 000 km above the surface of the Earth.
- They have an orbital period of 6 to 12 hours.
- These satellites stay in sight over a particular region of Earth for a longer time. The transmission distance and propagation delays are greater than those for LEO satellites.
- These orbits are generally polar in nature and are mainly used for communication and navigation applications.



- A **geosynchronous Earth orbit** is a prograde orbit whose orbital period is equal to Earth's rotational period.
- If such an orbit were in the plane of the equator and circular, it would remain stationary with respect to a given point on the Earth.

Satellite must fulfil the following conditions:

- It must have a constant latitude, which is possible only at 0° latitude.
- The orbit inclination should be zero.
- It should have a constant longitude and thus have a uniform angular velocity, which is possible when the orbit is circular.
- The orbital period should be equal to 23 hours 56 minutes, which implies that the satellite must orbit at a height of 35 786 km above the surface of the Earth.
- The satellite motion must be from west to east.

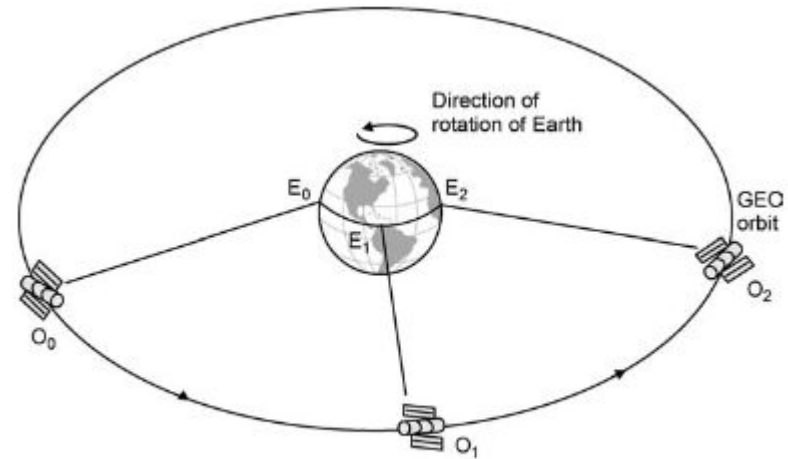


Figure 2.40 GEO satellites appear stationary wrt to a point on Earth



Sun-synchronous Orbit

Categorized as an LEO on the basis of distance from the surface of the Earth, needs a special mention and treatment because of its particular importance to satellites intended for remote sensing and military reconnaissance applications.

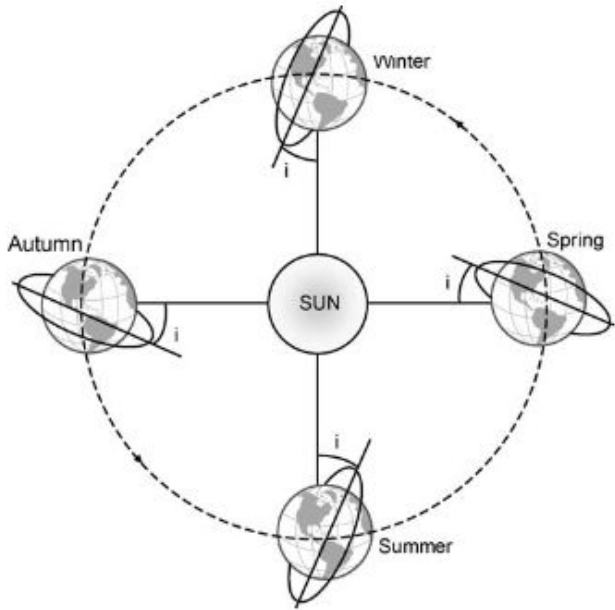


Figure 2.41 Sun-synchronous orbit

- A sun-synchronous orbit, is one that lies in a plane that maintains a fixed angle with respect to the Earth–sun direction.
- In other words, the orbital plane has a fixed orientation with respect to the Earth–sun direction and the angle between the orbital plane and the Earth–sun line remains constant throughout the year



Orbit Perturbations

- The ideal Keplerian orbit assumes the earth is a uniform spherical mass and that the only forces acting are the balancing gravitational pull and centrifugal force.
- In practice, other forces cause the orbit to deviate from this ideal path.
- The significant perturbing forces include:
 - Gravitational forces of the sun and the moon.
 - Atmospheric drag.
- The gravitational pulls of the sun and moon have a negligible effect on low-orbiting satellites, but they do affect satellites in the geostationary orbit.
- Atmospheric drag has a negligible effect on geostationary satellites but significantly affects low-orbiting earth satellites below about 1000 km.



Effects of a Nonspherical Earth (Oblateness)

The Oblate Spheroid Shape

- The earth is not perfectly spherical; it is an oblate spheroid, having an equatorial bulge and flattening at the poles.
- When this is accounted for, the mean motion (n) of the satellite is modified from the theoretical mean motion (n_0).
- The resulting orbital period is termed the anomalistic period (e.g., the time from perigee to perigee)



Rotations of the Orbital Plane

The earth's oblateness produces two main rotations of the orbital plane:

1. Regression of the Nodes

- The nodes appear to slide along the equator, rotating the line of nodes.
- The position of the right ascension of the ascending node (Ω) shifts.
- For prograde orbits, the nodes slide westward.
- For retrograde orbits, they slide eastward.
- For a polar orbit ($i=90^\circ$), the regression is zero.
- Sun-Synchronous Orbits are possible when orbital elements (a, e, i) are selected to yield an eastward regression rate of 0.9856° per day.

2. Rotation of Apsides

- The line of apsides rotates in the orbital plane, resulting in the argument of perigee (ω) changing with time.
- This rotation is zero when the inclination (i) is equal to 63.435° (a fact utilized in the orbits of Russian Molniya satellites).
- The overall result is that the satellite drifts (due to node regression) and the latitude of the point of closest approach (perigee) changes (due to apsides rotation)



Effects of a Nonspherical Earth (Equatorial Ellipticity)

- The earth's equator is not perfectly circular, exhibiting a small eccentricity ($\approx 10^{-5}$) known as equatorial ellipticity.
- This creates a gravity gradient that significantly impacts satellites in the geostationary orbit.
- The gradient causes GEO satellites to drift toward one of two stable points that coincide with the minor axis of the equatorial ellipse.
- These stable points, often referred to as "satellite graveyards," are located at approximately 75° E longitude and 105° W longitude.
- Station-keeping maneuvers must be performed to prevent GEO satellites from drifting to these points.
- The effect of equatorial ellipticity is negligible on most other satellite orbits



Atmospheric Drag

- Atmospheric drag is a significant force for near-earth satellites below about 1000 km.
- The drag is greatest at the perigee.
- This drag reduces the velocity at perigee, preventing the satellite from reaching its previous apogee height on subsequent revolutions.
- Consequences:
 - The semimajor axis (a) is reduced.
 - The eccentricity (e) is reduced.
 - Atmospheric drag does not noticeably change other orbital parameters, including perigee height.
- The resulting changes are typically significant only over long time intervals



Inclined Orbits

Introduction and Necessity of Coordinate Systems

- Studying a satellite in an inclined elliptical orbit is complex because different orbital parameters relate to different reference frames.
- Satellites in inclined orbits are not geostationary, meaning required look angles and range must change with time.
- Orbital elements are referenced to the orbital plane, which is fixed (or slowly varying) in space.
- The earth station location is given in local geographic coordinates, which rotate with the earth.
- Calculating satellite position and velocity requires rectangular coordinate systems.
- The overall calculation goal is finding the earth station look angles (azimuth and elevation) and range



Required Quantities and Coordinate Systems

- To determine look angles and range, the following are needed:
 - The orbital elements (from NASA bulletins, Sec. 2.6).
 - Various measures of time.
 - The Perifocal coordinate system (based on the orbital plane).
 - The Geocentric-equatorial coordinate system (based on the earth's equatorial plane).
 - The Topocentric-horizon coordinate system (based on the observer's horizon plane)



Antenna Look Angles

- The look angles for a ground station antenna are the azimuth (Az) and elevation (El) angles required for the antenna to point directly at the satellite.
- For a satellite in the geostationary orbit (GEO), calculating look angles is simpler than for an inclined elliptical orbit because the satellite appears stationary relative to the earth



Earth Eclipse of Satellite

I. Cause and Timing of Eclipses

- A geostationary satellite is eclipsed when it passes into the Earth's shadow.
- If the Earth's equatorial plane coincided exactly with the ecliptic plane (the plane of the Earth's orbit around the sun), geostationary satellites would be eclipsed by the Earth once every day.
- However, the equatorial plane is tilted at an angle of 23.4° to the ecliptic plane.
- This tilt allows the satellite to remain in full view of the sun for most days of the year.
- Eclipses occur when the satellite passes into the Earth's shadow. This occurs during periods around the:
 - Spring equinox (when the sun crosses the equator).
 - Autumnal equinox (when the sun crosses the equator).
- Daily eclipses begin approximately 23 days before the equinox and end approximately 23 days after the equinox for both the spring and autumnal periods



II. Duration and Operational Impact

- The duration of an eclipse varies over the period:
 - It lasts about 10 minutes at the beginning and end of the eclipse period.
 - It increases to a maximum duration of about 72 minutes at full eclipse.
- During an eclipse, the satellite's solar cells do not function.
- Storage batteries must be provided to supply operating power and maintain service



III. Longitudinal Preference

- The satellite's longitudinal position relative to the earth station affects when the eclipse occurs locally, which impacts operational usage.
- If the satellite longitude is east of the earth station, the satellite enters eclipse during daylight (and early evening) hours for the earth station.
 - This is undesirable because the satellite must operate on reduced battery power during these likely busy hours.
- If the satellite longitude is west of the earth station, eclipse does not occur until the earth station is in darkness (or early morning).
 - This is preferable because usage is likely to be low during this time.
- Therefore, satellite longitudes which are west, rather than east, of the earth station are more desirable for operational efficiency



Sun Transit Outage

- Sun Transit Outage is an event that must be accounted for during the periods around the equinoxes.
- It is caused by the transit of the satellite between the Earth and the Sun
- The outage occurs when the Sun comes within the beamwidth of the earth-station antenna
- When the sun transit occurs, the Sun acts as an extremely noisy source.
- This noise completely blanks out the signal from the satellite



Launch Vehicles

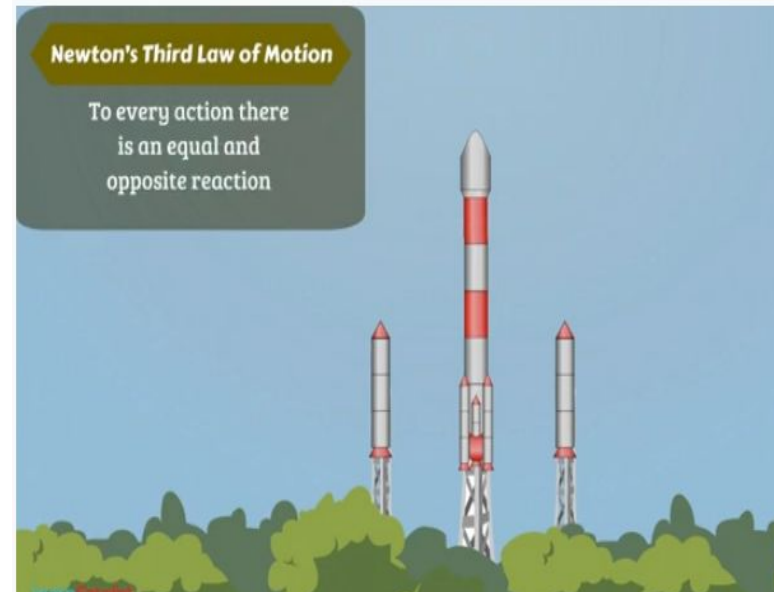
- A launch vehicle (also known as a rocket) is a vehicle used to propel spacecraft into space.
- It is designed to generate enough thrust to overcome Earth's gravity, reach orbit, or travel beyond the planet's atmosphere.
- The vehicle typically consists of multiple stages, each carrying fuel and having a separate engine that ignites in sequence.
- There are various types of propulsion systems used in launch vehicles, and each has distinct characteristics depending on the mission's needs.

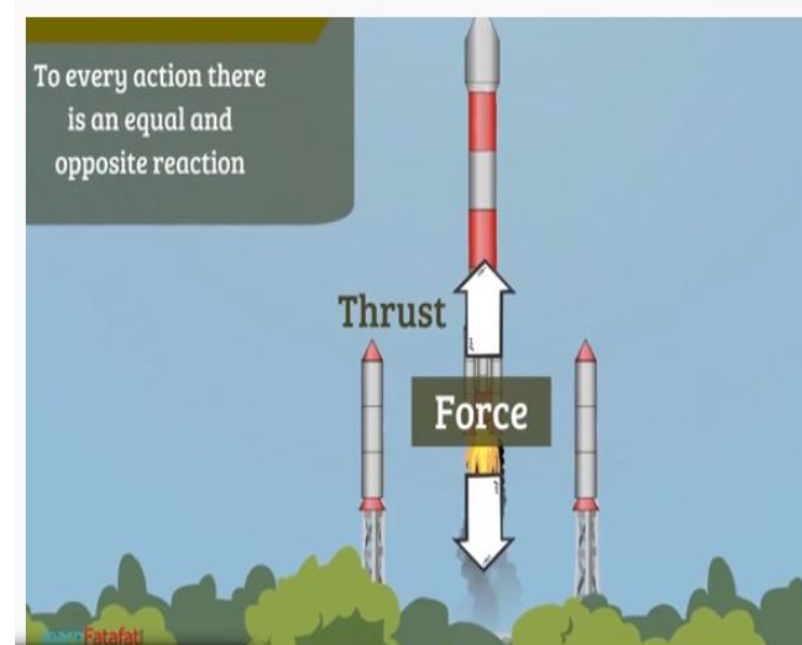
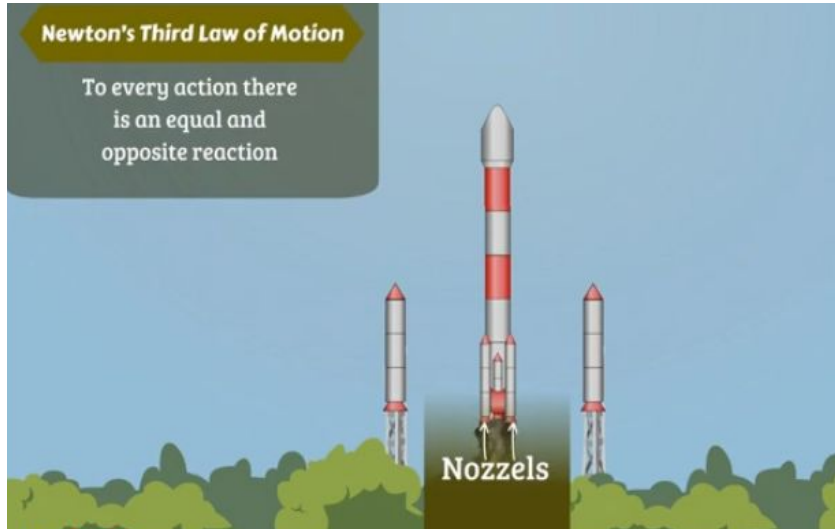


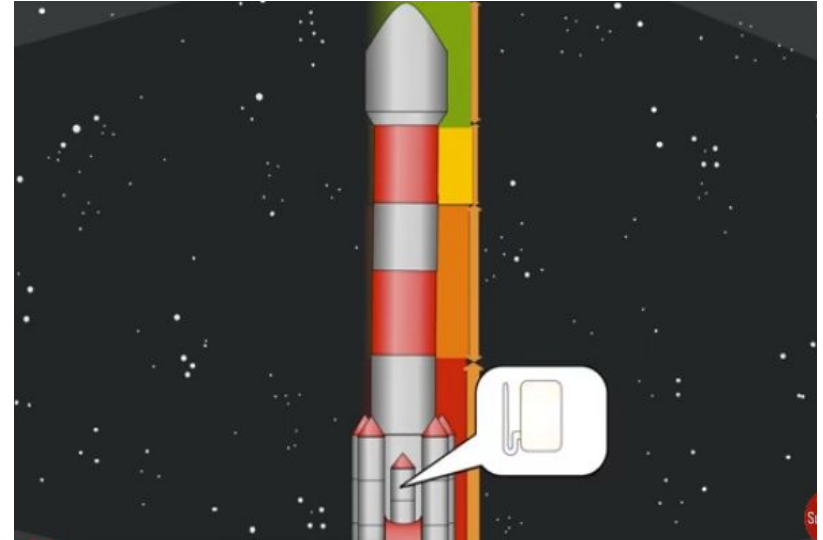
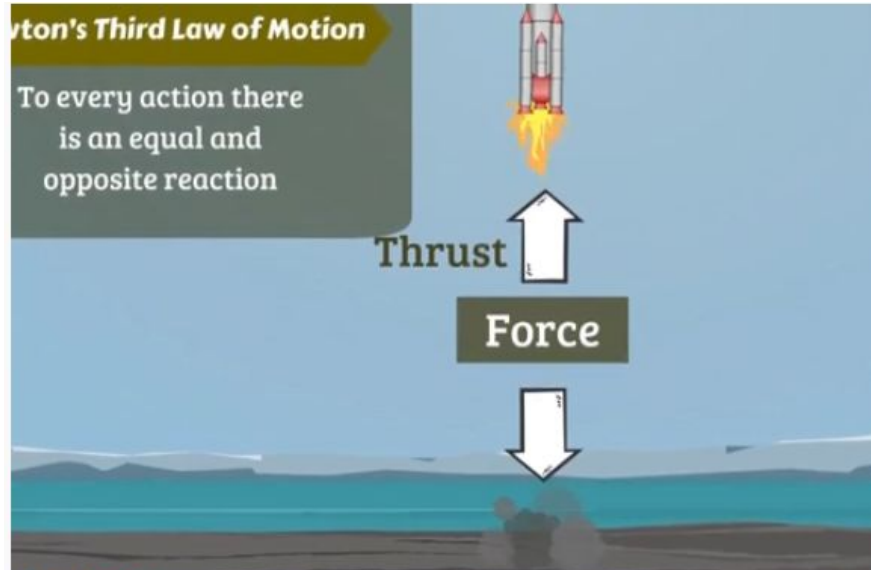
Types of LV

Expendable launch vehicles

Reusable launch vehicles













- A sounding rocket, used to study the atmosphere or perform brief experiments, is only capable of sub-orbital spaceflight and cannot reach orbit.
- A small-lift launch vehicle is capable of lifting up to 2,000 kg of payload into low Earth orbit (LEO).
- A medium-lift launch vehicle is capable of lifting between 2,000 to 20,000 kg of payload into LEO.
- A heavy-lift launch vehicle is capable of lifting between 20,000 to 50,000 kg of payload into LEO.
- A super-heavy lift vehicle is capable of lifting more than 50,000 kg of payload into LEO.



GSLV

- Geosynchronous Satellite Launch Vehicle abbreviated in English as GSLV, is an expendable launch system operated by the Indian Space Research Organization (ISRO).
- It was developed to enable India to launch its satellites without dependence on foreign rockets and providers.
- GSLV has been used in nine launches to date, since its first launch in 2001 through to its most recent launch on August 27, 2015 of the GSAT-6.



PSLV

- The Polar Satellite Launch Vehicle, commonly known by its abbreviation PSLV, is an expendable launch system developed and operated by the Indian Space Research Organization (ISRO).
- It was developed to allow India to launch its Indian Remote Sensing (IRS) satellites into Sun-synchronous orbits, a service that was, until the advent of the PSLV, commercially available only from Russia.
- PSLV can also launch small size satellites into geostationary transfer orbit (GTO).
- In the year 2015 alone India successfully launched 17 foreign satellites belonging to Canada, Indonesia, Singapore, United Kingdom and United States.
- Some notable payloads launched by PSLV include India's first lunar probe Chandrayaan-1, India's first interplanetary mission, Mangalyaan (Mars orbiter) and India's first space observatory, Astrosat



Rocketry

- A rocket is a spacecraft, aircraft, vehicle or projectile that obtains thrust from a rocket engine. Rocket engine exhaust is formed entirely from propellant
- A rocket is a machine that develops thrust by the rapid expulsion of matter.
- The basic components of a rocket include:
 - Payload: The object(s) being transported into space (e.g., satellites, cargo).
 - Propellant: The fuel and oxidizer combination that powers the rocket.
 - Structure: The frame that holds all components together.
 - Engine/Propulsion System: Provides the necessary thrust for launch.

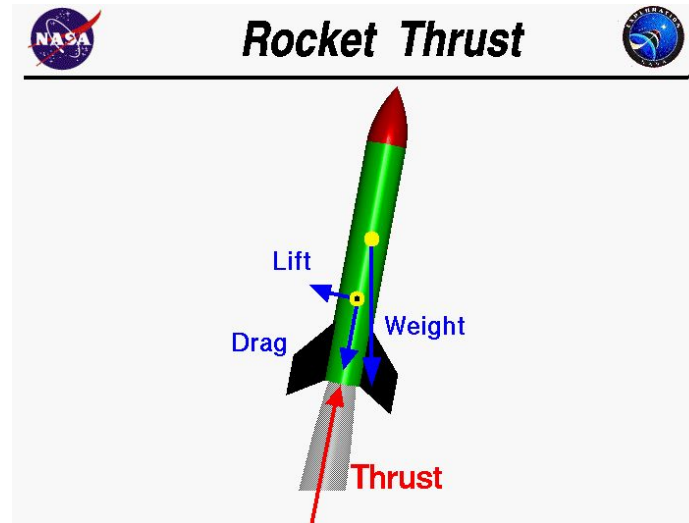


Terms of Rocket

- Thrust
- Impulse
- specific impulse
- Mass ratio



- **Thrust** is the force which moves the rocket through the air, and through space.
- Thrust is generated by the propulsion system of the rocket through the application of Newton's third law of motion; For every action there is an equal and opposite reaction





- **Specific impulse:** By definition, it is the total impulse (or change in momentum) delivered per unit of propellant consumed and is dimensionally equivalent to the generated thrust divided by the propellant mass flow rate or weight flow rate.
- Specific impulse is expressed in seconds. When the thrust and the flow rate remain constant throughout the burning of the propellant, the specific impulse is the time for which the rocket engine provides a thrust equal to the weight of the propellant consumed.



- In aerospace engineering, **mass ratio** is a measure of the efficiency of a rocket.
- It describes how much more massive the vehicle is with propellant than without; that is, the ratio of the rocket's wet mass (vehicle plus contents plus propellant) to its dry mass (vehicle plus contents).



LV stages





Types of Rocket Engines

- Chemical propellant rocket engines
- Ion rockets
- Photon rockets
- Magneto hydrodynamic
- Nuclear fission rockets



Chemical propellant rocket engines

- Liquid propellant
- Solid propellant
- Hybrid propellant



Liquid Propellant

Burns two separately stored liquid chemicals, a fuel and an oxidizer to produce thrust





FUEL	OXIDIZER	TYPE
Liquid Hydrogen	Liquid Oxygen	Cryogenic
RP-1 Kerosene	Liquid Oxygen	Liquid/Cryogenic
Aniline	Nitric Acid	Hypergolic
Hydrazine	Monopropellant	



Cryogenic Propellant

A cryogenic propellant is one that uses very cold, liquefied gases as the fuel and the oxidizer



Hypergolic Propellant

- A hypergolic propellant is composed of a fuel and oxidizer that ignite when they come into contact with each other.
- No spark is needed. Hypergolic propellants are typically corrosive so storage requires special containers and safety facilities



Mono- propellants

Monopropellants combine the properties of fuel and oxidizer in one chemical. By their nature, monopropellants are unstable and dangerous

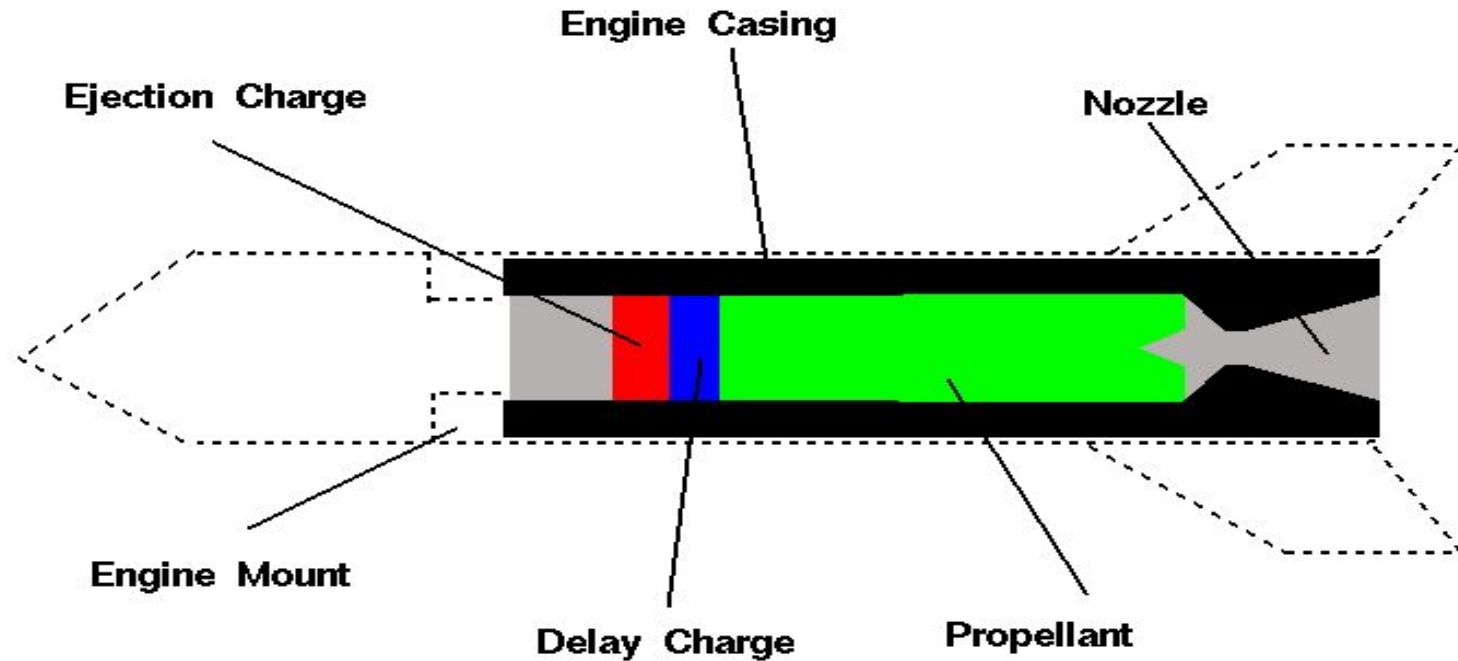


Solid Propellants

- Solid propellant rockets are basically combustion chamber tubes packed with a propellant that contains both fuel and oxidizer blended together uniformly.
- The principal advantage is that a solid propellant is relatively stable therefore it can be manufactured and stored for future use
- High density
- Burn very fast



Model Solid Rocket Engine





Hybrid Propellants

- The basic design of a hybrid consists of a combustion chamber tube, similar to ordinary solid fueled rockets, packed with a solid chemical, usually the fuel.
- Above the combustion chamber tube is a tank, containing a complementary reactive liquid chemical, usually the oxidizer



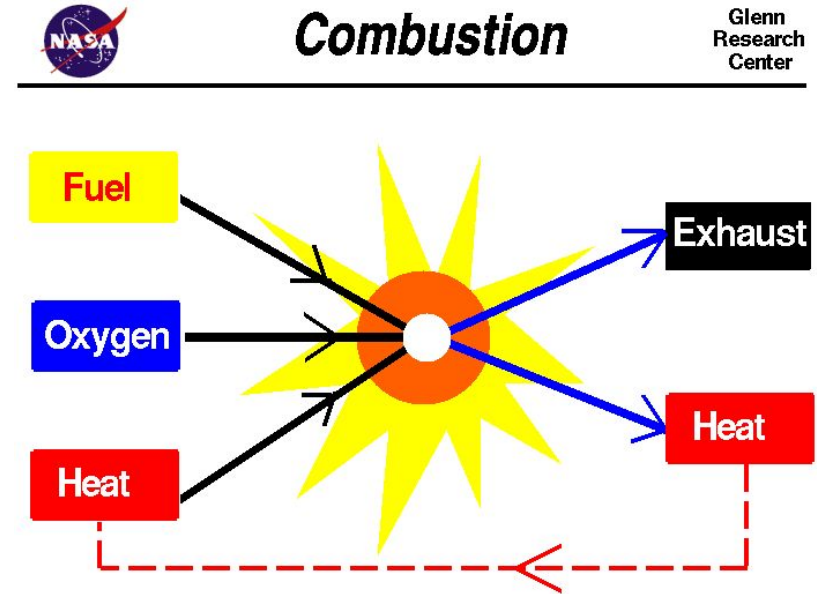
Propulsion

- Propulsion is a means of creating force leading to movement. The term is derived from two Latin words: pro, meaning before or forward; and pellere, meaning to drive.
- A propulsion system consists of a source of mechanical power, and a propulsion (means of converting this power into propulsive force).
- A technological system uses an engine or motor as the power source, and wheels and axles, propellers, or a propulsive nozzle to generate the force. Components such as clutches or gearboxes may be needed to connect the motor to axles, wheels, or propellers.
- Spacecraft propulsion is any method used to accelerate spacecraft and artificial satellites.
- propelled by forcing a gas from the back/rear of the vehicle at very high speed through a supersonic de Laval nozzle. This sort of engine is called a rocket engine



Combustion

- Combustion is a chemical process in which a substance reacts rapidly with oxygen and gives off heat. The original substance is called the fuel, and the source of oxygen is called the oxidizer.
- The fuel can be a solid, liquid, or gas, although for airplane propulsion the fuel is usually a liquid





Cryogenic Rocket Engine

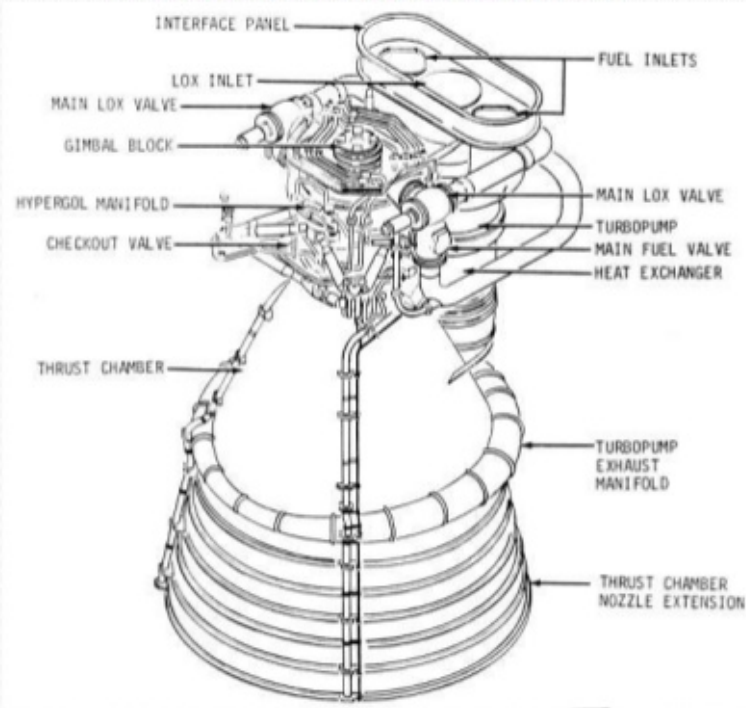
- A cryogenic rocket engine is a rocket engine that uses a cryogenic fuel or oxidizer, that is, its fuel or oxidizer (or both) are gases liquefied and stored at very low temperatures
- The combination of liquid hydrogen (LH₂) fuel and the liquid oxygen (LOX) oxidizer is one of the most widely used
- The major components of a cryogenic rocket engine are the combustion chamber (thrust chamber), pyrotechnic initiator, fuel injector, fuel cryo pumps, oxidizer cryo pumps, gas turbine, cryo valves, regulators, the fuel tanks, and rocket engine nozzle.
- In terms of feeding propellants to the combustion chamber, cryogenic rocket engines (or, generally, all liquid-propellant engines) are either pressure-fed or pump-fed, and pump-fed engines work in either a gas-generator cycle, a staged-combustion cycle, or an expander cycle.

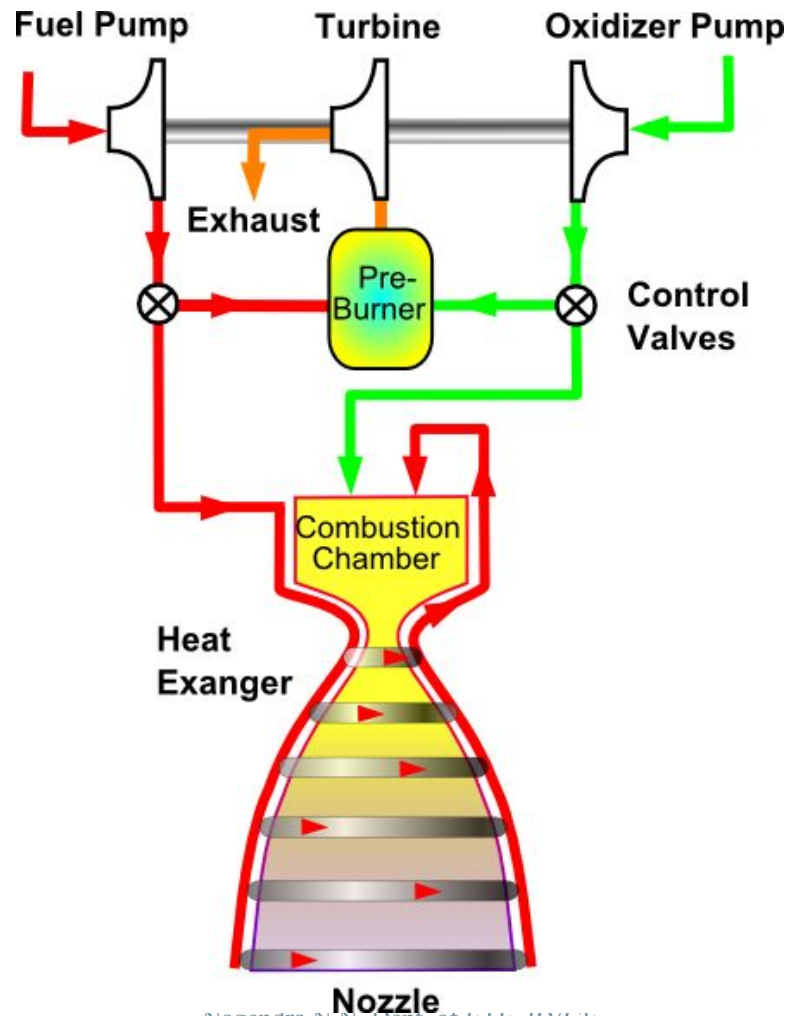


- The cryo-pumps are always turbo pumps powered by a flow of fuel through gas turbines.
- Looking at this aspect, engines can be differentiated into a main flow or a bypass flow configuration. In the main flow design, all the pumped fuel is fed through the gas turbines, and in the end injected to the combustion chamber



WHAT IS CRYOGENIC ENGINE...???

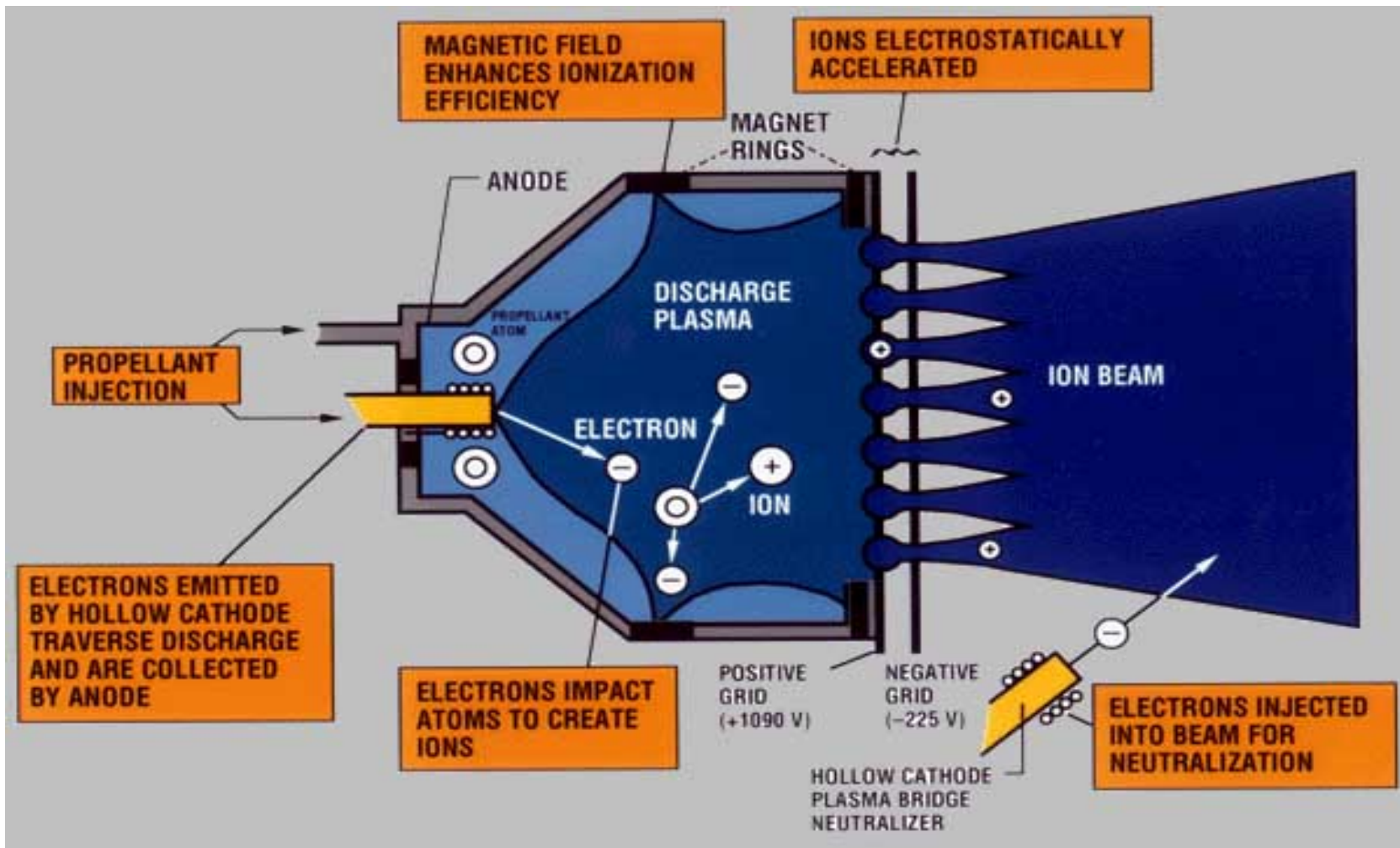






Ion propulsion

- An ion thruster is a form of electric propulsion used for spacecraft propulsion. It creates thrust by accelerating ions with electricity.
- Electrostatic
- Electromagnetic





Nuclear propulsion

- Nuclear propulsion includes a wide variety of propulsion methods that fulfill the promise of the Atomic Age by using some form of nuclear reaction as their primary power source.
- The propulsion plant of a nuclear-powered ship or submarine uses a nuclear reactor to generate heat. The heat comes from the fissioning of nuclear fuel contained within the reactor. Since the fissioning process also produces radiation, shields are placed around the reactor so that the crew is protected.



Pressurized-water Naval Nuclear Propulsion System

